Topology-Preserving Multi-Usage Pipe Centerline Refinement under Permanent Tubular Occupancy

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Abstract

Discrete multi-group pipe routers place channels on a grid or coarse graph. The resulting centerlines are topologically feasible but typically contain many axis-aligned corners. For permanently occupied fluid tubes—as in additively manufactured hydraulic manifolds—those corners are undesirable: they degrade geometric fairness and, in practice, flow quality. This technical report studies the downstream geometric stage that starts from a fixed routing topology and a feasible coarse layout, and refines tubular centerlines under permanent clearance.

Unlike classical robotic path smoothing, which usually fair one temporary open trajectory, our setting is a branched manifold with declared terminal usages, finite tube radii, and static cross-group occupancy. We formalize the refinement problem, highlight the multi-usage junction conflict that a single shared branch point cannot resolve, and outline a hybrid trunk-share / junction-split parameterization with soft geometric costs and hard clearance acceptance. On a multi-group suite (8 groups $\times$ 8 terminals, r=2) with capsule clearance, the refine reaches full hard-accept success, $\kappa r \lesssim 1$, and about a 95% relative drop in the soft objective. Physics-based pressure-drop optimization is intentionally deferred to a later PDE stage; here the objective remains geometric.

Code. Reference implementation, instance loaders, and batch runners are in `https://github.com/bladesaber/MAPF_Pipeline` (branch `branch_quan`), under `pipe_smooth/`.

1 Introduction

Additive manufacturing (AM) allows compact hydraulic components whose internal channels follow free-form three-dimensional paths rather than intersecting drilled holes. A representative design case is multi-group channel layout in rotary direct-drive servo valves: terminals are partitioned into groups; pipes of different groups must remain geometrically separated for all poses of interest, while pipes within a group may share trunks and branch into tree-like manifolds. Once a channel is placed it permanently occupies volume, so later groups must clear earlier geometry.

Our companion work on discrete multi-group pipe routing under permanent geometric occupancy Deng [2025] addresses the combinatorial layout problem: given ports, obstacles, and group topologies, produce a conflict-free coarse embedding on a constructive 3D grid benchmark, evaluated by fixed-budget success of CBS, PBS, and priority planning. That paper explicitly leaves post-layout smoothing as orthogonal follow-on work. The layouts it returns are feasible but still grid-angular; the present report refines those centerlines.

Why robotic path smoothing is not enough. Trajectory fairing methods such as CHOMP, STOMP, TEB, and TrajOpt typically optimize a single open curve for a temporary agent under time-indexed or soft obstacle costs Zucker et al. [2013], Kalakrishnan et al. [2011], Rösmann et al.

[2012], Schulman et al. [2014]. Pipe refinement differs along several axes that we make precise in Sec. 2: permanent tubular occupancy, branched topology with degree-≥ 3 junctions, costs driven by declared usages rather than one s–t path. The central geometric conflict is not "make one spline smoother," but *how several overlapping usages share a trunk while remaining fair at every declared terminal-to-terminal flow.*

Scope of this report. We formalize topology-preserving, multi-usage geometric refinement under permanent tubular occupancy (Sec. 2), survey adjacent literatures (Sec. 3), and describe a hybrid trunk-share / junction-split Hermite parameterization (Sec. 4) with soft path and clearance costs (Sec. 5). Gradient-based experiments on synthetic instances with capsule clearance follow in later sections. We do not claim a new optimal continuous planner, commercial CAD parity, or CFD-validated hydraulic improvement; those belong to later CAD and fluid optimization, not to this geometric pass.

Contributions.

1. **Problem.** A clear formulation of post-routing centerline refinement for branched, multi-usage, permanently occupied tubes, including the shared-junction versus locally split junction regimes.
2. **Method sketch.** Hybrid trunk-share / junction-split variables with soft length/curvature and clearance costs, hard ports and topology by construction, and hard clearance at accept (detailed in later sections).
3. **Empirical note.** On hard multi-group synthetic cases, gradient refine with capsule clearance yields feasible designs with $\kappa r_{\max} \lesssim 1$ and large relative objective drop (Sec. 6).

2 Problem Formulation

2.1 Setting

The geometric refine starts after a discrete multi-group router has returned a feasible coarse layout. We treat the following as *fixed*:

- **Domain** Ω: a bounded design region with static obstacles and already committed pipes of other groups (permanent clearance).
- **Topology** $\mathcal{T}$: for each group, a connected graph (typically a tree), together with a feasible coarse polyline or grid embedding that initializes the refine.
- **Usages** $\mathcal{U}$: a set of directed terminal–terminal flows that must be considered when scoring fairness (e.g. $A \to C$, $B \to C$, $A \to B$).
- **Radii and ports**: tube radii along the layout, fixed terminal positions, and optional end-tangent / line-axis clamps so that a port sits on a prescribed axis.

The decision variables X parameterize centerlines (and, implicitly, tubular solids of the given radii). The output must keep every terminal connected as in $\mathcal{T}$: each declared usage remains a continuous path between its ends. Branches need not meet at a single geometric point; Sec. 2.4 explains when a local split near a junction is allowed.

2.2 Objective sketch

Given topology $\mathcal{T}$ and a feasible coarse embedding, refine the geometric parameters X so that a weighted soft cost over declared usages is minimized, subject to hard geometric constraints:

$$\min_X \; J_{\text{path}}(X) + J_{\text{clear}}(X) + J_{\text{union}}(X) + J_{\text{space}}(X) \quad \text{s.t. ports, } \mathcal{T}\text{, interface } C^0/C^1 \text{ by construction.} \tag{1}$$

Table 1 summarizes the roles of the terms. Full discrete definitions appear in Sec. 5; here we only fix the semantics needed for the problem statement.

Table 1: Soft versus hard elements in the geometric refine.

Symbol	Role	Soft / hard
X	Centerline parameters (shared trunk + private junction controls)	decision
L, K in J_{path}	Chord energy (each shared piece once) + bend (segment interiors once; usage joints once)	soft
J_{clear}	Obstacle and cross-group tube separation	soft in loop; **hard** at accept
J_{union}	Junction-bundle compactness	soft
J_{space}	Keep sample gaps comparable under parameter sampling	soft
Topology / ports / interface	Connectivity, fixed ends, C^0/C^1 clamps	**hard** (by construction)

2.3 Difference from classical path smoothing

Table 2: Classical path smoothing versus the pipe refine studied here.

Aspect	Normal path smoothing	This problem
Occupancy	Pass-and-release (time / moving agent)	Permanent tubes; clearance is static geometry
Structure	Single s–t line	Tree / branched manifold; degree ≥ 3 joins
Cost scope	One path's length / curvature	Usage-assembled paths over the branched manifold
Domain	Often open or mild	Tight Ω may force large κ (soft K)
Ends	Fixed s, t (pose optional)	Fixed ports plus line-axis alignment

2.4 Junction conflict: shared node versus local split

Usages matter because shared trunk nodes couple directions. Optimizing only the shared polyline can make $A \to C$ and $B \to C$ smooth while $A \to B$ remains a hairpin at the branch. The declared set $\mathcal{U}$ therefore decides which end-to-end paths must stay fair, and where junctions need private degrees of freedom.

Consider a Y-junction with fixed topology and usages $\mathcal{U} = \{A \to C,\ B \to C,\ A \to B\}$:

$$A\text{—}J\text{—}C, \qquad B\text{—}J.$$

Shared junction. All three branches meet at one point J. Fair geometry for $A \to C$ and $B \to C$ forces a sharp turn on $A \to B$ at J (Fig. 1a).

Locally split junction. Private junction degrees of freedom let $A \to B$ take a dedicated arc while $A \to C$ and $B \to C$ keep a shared trunk (Fig. 1b). A void appears near the join; connectivity of $\mathcal{T}$ is preserved, but the embedding need not keep a common junction point.

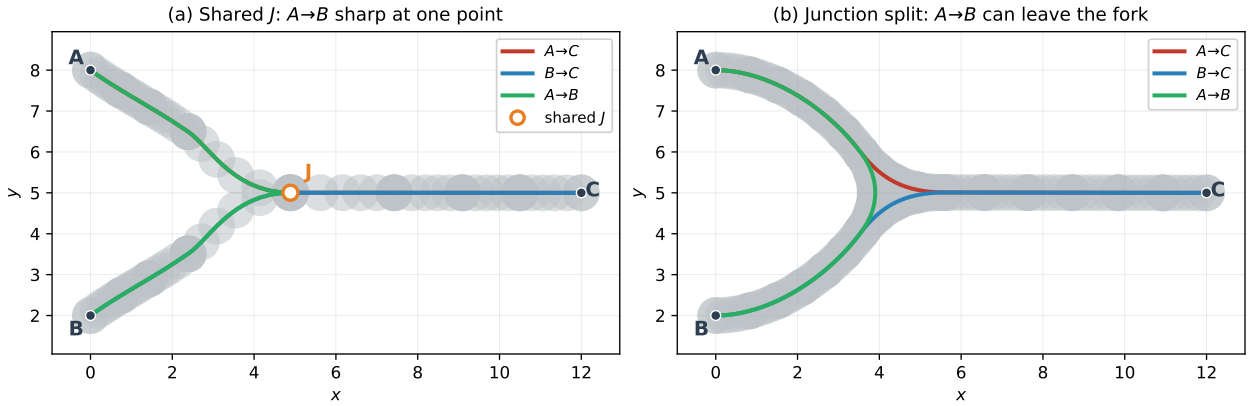


Figure 1: Planar Y under the same usages $\mathcal{U} = \{A \to C,\ B \to C,\ A \to B\}$ (top view; gray disks = tube radius). (a) Forced shared connect point: the three usage centerlines meet at one node J (orange); refine can fair $A \to C$ and $B \to C$, but $A \to B$ (green) stays sharp at J. (b) Hybrid trunk-share / junction-split: $A \to B$ can leave the fork while the trunk toward C remains shared.

Table 3: Shared junction versus locally split junction.

	Shared junction	Locally split junction
Geometry at join	one shared point J	trunks + dedicated $A \to B$ arc + void
$A \to C$, $B \to C$	arc + straight	same as shared case
$A \to B$	sharp at J	dedicated arc; fair
$\mathcal{T}$	preserved	preserved (connectivity, not a common join point)

The design implication is that the refine must allow private degrees of freedom near junctions (local split) while sharing trunk controls where usages truly share physical pipe. That hybrid structure—trunk-share + junction-split—is the method focus of this report. The core gap relative to single-curve fairing is therefore **topology-preserving, multi-usage geometric refinement under permanent tubular occupancy**, not merely denser spline fitting.

2.5 Place in the pipeline

A typical valve-channel workflow is discrete multi-group routing, then centerline refine, then CAD solids, then fluid PDE / shape polish. This report covers only the geometric refine.

That ordering matters for junctions. Downstream shape or adjoint PDE optimization usually deforms a *fixed* channel layout: it can move walls or a shared centerline, but it does not introduce a new split curve or triangular void at a Y-join the way a locally split junction does. That local split is therefore a decision of the geometric refine, not of later physics polish. We still preserve the connectivity of $\mathcal{T}$ (no discrete rewiring); what changes is the geometric arrangement near degree-≥ 3 nodes. Pressure drop and streamline quality remain for subsequent physics-based optimization, once the centerline graph—including any local splits—is fixed.

3 Related Work

Discrete pipe routing. Belov et al. Belov et al. [2020] adapt MAPF-style search to 3D pipe routing; surveys of plant-scale automatic routing emphasize sequential CAD and MIP/sampling pipelines Blokland et al. [2023]. Our companion report Deng [2025] treats *multi-group* routing under permanent occupancy and supplies the coarse layouts we refine. That work stops at discrete embeddings; continuous centerline cleanup is left open.

Robotic path smoothing. CHOMP, STOMP, TEB, and TrajOpt Zucker et al. [2013], Kalakrishnan et al. [2011], Rösmann et al. [2012], Schulman et al. [2014] refine a single temporary trajectory under smoothness and collision costs. We inherit the gradient-centric idea, but a one-path objective does not encode shared trunks, permanent cross-group tubes, or the multi-usage junction conflict of Sec. 2.4.

Fairing, CAD, and physics polish. Elastic-rod / curve fairing Moreton and Séquin [1992], Bergou et al. [2008] and clothoid primitives Fraichard and Scheuer [2004] motivate soft bend energies on one open curve; sparse circular-arc encodings are a poor fit for dense grid-routed manifolds. Parametric Hermite / NURBS centerlines are standard in CAD Piegl and Tiller [1997], yet manual fillets do not resolve multi-usage joins under hard clearance. Adjoint CFD / shape optimization sits *downstream*: it typically deforms a fixed channel layout and does not create a locally split junction (Sec. 2.5). Our contribution is the combination—branched, usage-coupled refine under permanent tubular occupancy—not any single fairing or collision ingredient.

4 Method

This section fixes the parameterization and the optimize loop; cost formulas are in Sec. 5.

4.1 Pipeline overview

For each group independently (cross-group coupling enters only through clearance):

1. **Tree.** Deduplicate the coarse grid / polyline embedding into an undirected tree $\mathcal{T}$ (or forest): nodes are terminals and branch points; edges carry the coarse polylines and radii.

2. **Junction partition.** Grow junction complexes around degree-≥ 3 nodes; label remaining edges as trunks (Sec. 4.2).

3. **Hermite graph.** Instantiate cubic Hermite segments on labeled edges. Trunk knots are shared; junction knots are private per labeled junction edge; ports and trunk–junction interfaces are clamped for C^0/C^1 by construction (Sec. 4.3).

4. **Assemble & optimize.** Sample usages, minimize soft J by gradient steps, accept under hard clearance (Alg. 1).

Groups couple only through J_{clear} and the final accept test. Topology $\mathcal{T}$ and $\mathcal{U}$ are never rewritten.

4.2 Junction partition: where DOF are private

The hybrid rule is: *share controls wherever usages share physical pipe; add private controls only where multi-usage curvature conflicts.*

Grow complexes. Seed every tree node of degree ≥ 3. Grow a zone around each seed by tree-path length with threshold $\tau = \max(\tau_0, \alpha\, r_{\text{local}}, \beta\, R_{\text{bend}})$. After the zone grow, any connected trunk component whose total length is shorter than a threshold τ_{trunk} is absorbed into junction (so nearby seeds separated by a short trunk become one complex Z). Edges inside Z are labeled `junction`; edges outside all complexes are `trunk`.

Variable layout. Table 4 summarizes roles after the label pass. When building the Hermite graph we apply one privatization rule: *ports and complex interfaces keep a single global knot; every other sample on a labeled edge is private to that edge*, so two junction edges that meet at the same coarse branch point do not share interior (p, v). Usages do not each own a full duplicate spline; they inherit geometry by walking the labeled edges on their tree path. Junction chains stay short, so new freedom appears only near the conflict in Fig. 1.

Table 4: Region roles after junction partition.

Region	Geometry	Variables	Continuity
Trunk	Edges outside all complexes	Shared Hermite knots X_{trunk}	Intrinsic (one curve)
Junction	Interior of complex Z	Private knots *per labeled junction edge*	Meets trunk at interface
Interface	Complex boundary	Shared endpoints (+ clamped tangents)	Hard C^0; hard C^1 by construction
Ports	Group terminals	Fixed position / axis tangent	Hard clamps

Y example (how a local split appears). On A–J–C with B–J and $\mathcal{U} = \{A{\to}C,\, B{\to}C,\, A{\to}B\}$, the coarse embedding places all three branches at one point J. After partition, each usage is a short alternating chain of *shared trunks* and *private junction edges*:

$$
\begin{aligned}
A{\to}C:&\ A_{\text{trunk}} \ \to\ AC_{\text{junc}} \ \to\ C_{\text{trunk}},\\
B{\to}C:&\ B_{\text{trunk}} \ \to\ BC_{\text{junc}} \ \to\ C_{\text{trunk}},\\
A{\to}B:&\ A_{\text{trunk}} \ \to\ AB_{\text{junc}} \ \to\ B_{\text{trunk}}.
\end{aligned}
$$

Here A_{trunk}, B_{trunk}, and C_{trunk} are shared Hermite geometry (every usage that enters that arm reads the same knots). The three junction edges AC_{junc}, BC_{junc}, and AB_{junc} are independent: each has its own interior (p, v), even though they all start at the same coarse coordinates near J. Thus $A \to C$ and $B \to C$ couple through C_{trunk}, while $A \to B$ can fair through AB_{junc} without forcing a hairpin at a single shared point. Initialization copies the shared-junction layout, so the private junction knots begin coincident; during optimization they may separate and open the void in Fig. 1b—a continuous motion of already-private DOF, not a discrete mode switch.

4.3 Cubic Hermite representation

Each labeled edge is a chain of cubic Hermite segments. A segment with ends (p_0, v_0) and (p_1, v_1) is, for $t \in [0, 1]$,

$$\gamma(t) = H_{00}(t)\, p_0 + H_{10}(t)\, v_0 + H_{01}(t)\, p_1 + H_{11}(t)\, v_1, \tag{2}$$

with the standard Hermite basis

$$\begin{aligned} H_{00}(t) &= 2t^3 - 3t^2 + 1, & H_{10}(t) &= t^3 - 2t^2 + t, \\ H_{01}(t) &= -2t^3 + 3t^2, & H_{11}(t) &= t^3 - t^2. \end{aligned}$$

Thus $\gamma(0) = p_0$, $\gamma(1) = p_1$, $\gamma'(0) = v_0$, $\gamma'(1) = v_1$. Shared nodes enforce hard C^0 and C^1: every incident segment reads the same (p, v).

Clamps. Port nodes fix p and align v with the prescribed unit vector. Interface nodes between trunk and junction share p (and, by construction, compatible v) so that usage walks stay G^1 across the complex boundary. Only movable trunk knots and private junction knots enter the decision vector x.

4.4 Assemble and optimize

Alg. 1 minimizes the soft objective of Sec. 5 by first-order steps on $\nabla_x J$ and accepts a design only under hard obstacle / cross-group clearance. Cost formulas are in Sec. 5; which first-order variant we use in practice is an implementation detail reported with the experiments.

Algorithm 1 Usage assembly and gradient refine (per multi-group instance)

Require: Hermite graphs $\{G_g\}$, usages $\mathcal{U}$, weights, steps T, refresh period k

$x \leftarrow$ concatenate free node positions and tangents (p, v) across groups
for $t = 1, \ldots, T$ **do**
    write x into the Hermite graphs
    if $t = 1$ or $(t-1) \bmod k = 0$ **then**
        find candidate colliding pairs by proximity (broadphase):
    $\mathcal{N}_{\text{obs}}$: tube samples vs obstacle primitives
    $\mathcal{N}_{\text{grp}}$: tube samples of group g vs group g'
    $\mathcal{N}_{\text{union}}$: nearby junction samples within a complex
    end if
    for each usage $u \in \mathcal{U}$ **do**
        follow u's path on $\mathcal{T}$ (shared trunks + private junctions)
        sample Hermite segments along that path $\rightarrow \gamma_u$, r_u
    end for
    $J_{\text{path}} \leftarrow$ length / curvature on $\{\gamma_u, r_u\}$ ▷ Sec. 5
    $J_{\text{space}} \leftarrow$ spacing penalty on chord gaps
    $J_{\text{clear}} \leftarrow$ soft gaps on pairs in $\mathcal{N}_{\text{obs}} \cup \mathcal{N}_{\text{grp}}$
    $J_{\text{union}} \leftarrow$ soft compactness on pairs in $\mathcal{N}_{\text{union}}$
    $J \leftarrow J_{\text{path}} + J_{\text{space}} + J_{\text{clear}} + J_{\text{union}}$
    differentiate J w.r.t. x through the sampling map (reverse-mode AD)
    $x \leftarrow$ gradient-based update using $\nabla_x J$
end for
write x back; hard-check obstacle / cross-group clearance (no soft margin)
accept iff hard hit count = 0; else reject / raise clearance weights

Ensure: accepted Hermite graphs, or failure

5 Cost Functions

Alg. 1 minimizes a scalar soft objective

$$J(x) = J_{\text{path}}(x) + J_{\text{clear}}(x) + J_{\text{union}}(x) + J_{\text{space}}(x). \tag{3}$$

Terms below are evaluated on Hermite samples (and proximity pairs from Alg. 1).

5.1 Path costs

We set

$$J_{\text{path}} = \alpha\, L + \beta\, K. \tag{4}$$

Let $\mathcal{E}$ be the set of Hermite segments (trunk and junction) and $\mathcal{J}$ the set of unordered segment–segment joints discovered by walking usages.

Aggregation (not a naive usage sum). Chord energy charges each physical segment once,

$$L = \sum_{e \in \mathcal{E}} L_e, \tag{5}$$

so a shared trunk is not multiplied by how many usages traverse it. Curvature sums segment interiors once and every distinct usage joint once,

$$K = \sum_{e \in \mathcal{E}} K_e^{\text{int}} + \sum_{\{e,e'\} \in \mathcal{J}} K_{e,e'}^{\text{joint}}. \tag{6}$$

Walking $\mathcal{U}$ is still required for K: the hairpin at a shared Y lives at *which* segments meet, even when trunk interiors were already scored. A naive $\sum_u L_u$ (or $\sum_u K_u$ over full usage polylines) would over-count shared material and over-pull trunks.

Chord energy. On a sample chain $\{x_i\}$, we use squared chords rather than $\sum_i \|x_{i+1} - x_i\|_2$:

$$L_e = \sum_i \|x_{i+1} - x_i\|_2^2 \tag{7}$$

(with optional sample weights $\bar{w}_i = \frac{1}{2}(w_i + w_{i+1})$). Squaring penalizes uneven gap distributions more strongly than raw chord length (one long chord costs more than several short ones of equal total length).

Radius-aware bend. The same centerline turn is worse for a fatter tube, so we penalize a dimensionless bend related to κr, not bare κ. With unit chords t_i and turning angle

$$\theta_i = \operatorname{atan2}\big(\|t_{i-1} \times t_i\|_2,\ t_{i-1} \cdot t_i\big) \in [0, \pi] \tag{8}$$

(preferred over $\arccos(t_{i-1} \cdot t_i)$, which is ill-behaved as $\theta \to 0$), the bend energy on a chain is

$$\sum_i \frac{(\theta_i\, r_i)^2}{\Delta \ell_i + \varepsilon} \approx \int (\kappa r)^2 \, \mathrm{d}s, \tag{9}$$

with chord lengths $\Delta s_i = \|x_{i+1} - x_i\|_2$ and $\Delta \ell_i = \frac{1}{2}(\Delta s_{i-1} + \Delta s_i)$. The same kernel is applied on segment interiors (K_e^{int}) and on three-point windows at usage joints ($K_{e,e'}^{\text{joint}}$).

5.2 Clearance

For two capsule / ball witnesses with centers a, b and radii r_a, r_b, let $d_{\text{req}} = r_a + r_b + \delta_{\text{margin}}$ and

$$\phi(a, b; r_a, r_b) = \Big(\big[d_{\text{req}} - \|a - b\|_2 \big]_+^{\sigma} \Big)^2, \tag{10}$$

where $[\cdot]_+^{\sigma}$ is a softplus hinge. Then

$$J_{\text{clear}} = J_{\text{obs}} + J_{\text{grp}}, \tag{11}$$

with J_{obs} summing ϕ over tube–obstacle pairs in $\mathcal{N}_{\text{obs}}$ and J_{grp} summing ϕ over cross-group pairs in $\mathcal{N}_{\text{grp}}$ (Alg. 1). Weights μ_{obs} and μ_{grp} scale the two sums.

Hard accept. After the loop we recompute residuals on obstacle / cross-group pairs *without* the soft margin used inside J_{clear} and accept only if the hard hit count is zero (Alg. 1). Capsule proxies are incomplete relative to exact swept solids, but sufficient for the geometric refine before CAD.

5.3 Junction union

Private junction patches are under-determined by L and K alone. Inside each complex Z, let γ_u^Z be the samples of usage u on junction edges of Z, and let $\mathcal{U}_Z$ be the usages that enter Z. For a point x with radius r_x and a curve γ with nearest-sample radius $r_\star$, set $r_\Sigma = r_x + r_\star$ and

$$\psi(d; r_\Sigma) = \begin{cases} \big(d/r_\Sigma\big)^2 & d < \alpha\, r_\Sigma, \\ 0 & \text{otherwise}, \end{cases} \tag{12}$$

with cutoff factor $\alpha \sim O(1)$. The pairwise Chamfer

$$\mathrm{Ch}(\gamma_u^Z, \gamma_v^Z) = \sum_{x \in \gamma_u^Z} \psi\big(d(x, \gamma_v^Z); r_\Sigma\big)\, \Delta s(x) \;+\; \sum_{y \in \gamma_v^Z} \psi\big(d(y, \gamma_u^Z); r_\Sigma\big)\, \Delta s(y) \tag{13}$$

then defines

$$J_{\text{union}} = \lambda \sum_Z \sum_{\{u,v\} \subset \mathcal{U}_Z} \mathrm{Ch}(\gamma_u^Z, \gamma_v^Z). \tag{14}$$

While usages remain within about one radius sum, ψ pulls them into a compact bundle; once they separate beyond the cutoff (local split, Fig. 1b), the force releases. Radius normalization keeps λ comparable across tube sizes.

5.4 Spacing regularizer

Hermite segments are sampled in the parameter $t \in [0, 1]$, not in arc length. Without a spacing term, the optimizer can stretch $\|\gamma'(t)\|$ so that large gaps open between samples: soft clearance and bend then see only a sparse, biased subset of the curve. We therefore use

$$J_{\text{space}} = \eta \sum_i \big(\Delta s_i - \overline{\Delta s}\big)^2, \tag{15}$$

where $\overline{\Delta s}$ is the mean chord on the chain, to keep sample gaps comparable and reduce this parameterization artifact.

6 Experiments

We evaluate the hybrid refine on a synthetic multi-group suite with capsule clearance. The goal is to check that the method is usable at scale: hard clearance accept, dimensionless bend $\kappa r \lesssim 1$, and a clearly decreasing soft objective.

6.1 Setup

Instances. We use N=50 instances from `hard_r2`: each is a 48^3 grid layout from the companion discrete router Deng [2025] (PlaneSlice corridors + PillarField obstacles), with **8 groups**, **8 terminals per group**, and uniform tube radius r=2. Coarse paths and declared usages are fixed inputs; the refine does not replan topology.

Protocol. All runs use capsule clearance, Chamfer junction union, and the costs of Sec. 5. Free Hermite DOF are updated for a budget of 1500 first-order steps (Alg. 1) with step size 0.01, soft clearance margin 0.15, and proximity-pair refresh every 10 steps. Weights are α=1, β=1, μ_{obs}=1, μ_{grp}=1, λ=0.1, η=10.

Metrics.

- **Success:** fraction of instances with hard hit count $n_{\text{hit}} = 0$ (obstacle and cross-group; no soft margin).
- **Path:** aggregate true polyline length L, max dimensionless bend $\kappa r_{\max}$, and voxel tube volume V (accepted runs).
- **Solve:** wall time, final soft objective J_T, and relative drop $(J_0 - J_T)/J_0$.

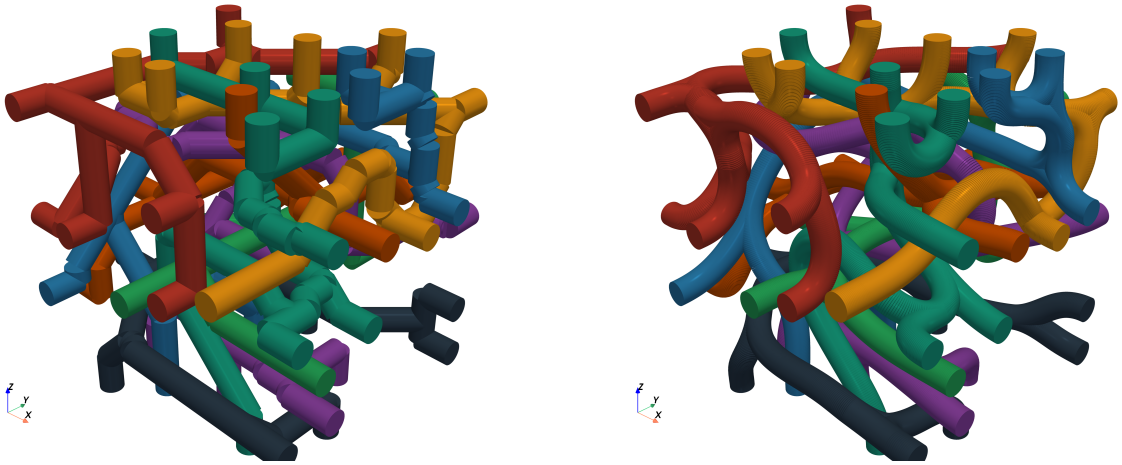
6.2 Results

Table 5 reports the batch. All 50 instances are accepted. Mean $\kappa r_{\max} \approx 0.97$ stays below one, and the soft objective drops by about 95% on average ($J_0 \sim 6.55 \times 10^3 \to J_T \sim 3.60 \times 10^2$). Mean wall time is about 160 s per instance under this protocol.

Table 5: Hard_r2 refine with capsule clearance (N=50). Means over all runs (all accepted).

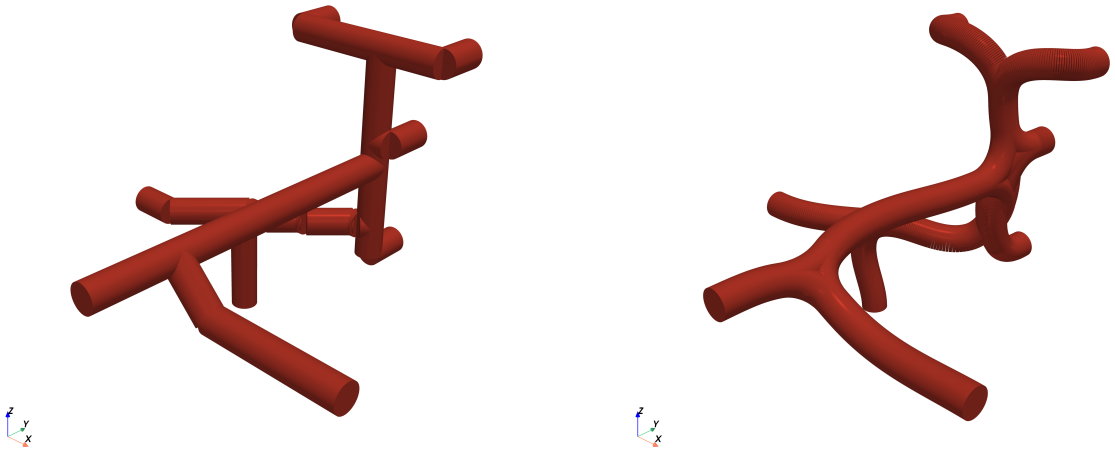
Success	Time (s)	J_T	Rel. drop	L	$\kappa r_{\max}$	V
50/50	160	360	0.95	4298	0.97	1.91×10^4

Fig. 2 shows the full multi-group packing of instance 0 before and after refine (colors mark groups; obstacles omitted for clarity). Fig. 3 isolates one group so that grid corners vs. fair Hermite tubes are easier to read. Fig. 4 plots the soft objective for that run; J drops sharply in the first few hundred steps and then plateaus.



Before (coarse) After (refined)

Figure 2: All groups of instance 0 from `hard_r2`: coarse discrete paths (left) vs. refined Hermite tubes (right). Colors mark groups; obstacles omitted.



Before (coarse) After (refined)

Figure 3: Single group (g_2) of the same instance, obstacles omitted: coarse discrete paths (left) vs. refined Hermite tubes (right).

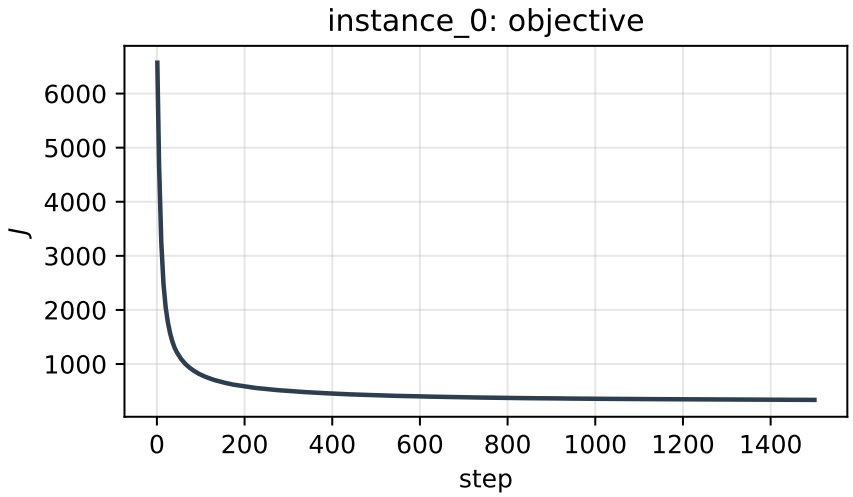


Figure 4: Soft objective J vs. iteration for instance 0 (`hard_r2_w1` / Adam / capsule).

7 Conclusion

We formalized topology-preserving, multi-usage pipe centerline refinement under permanent tubular occupancy, and instantiated it with hybrid trunk-share / junction-split Hermite variables, soft path / clearance / union / spacing costs, and hard clearance at accept. On 50 hard multi-group synthetic instances with capsule clearance, the refine reaches full hard-accept success, $\kappa r_{\max} \lesssim 1$, and about a 95% relative drop in the soft objective.

The objective here is geometric: K is a bend proxy, not pressure drop, and capsule clearance is a fast narrowphase rather than exact CAD interference. Fluid PDE polish and automatic weight schedules are left to later work; this report only claims a usable geometric stage between discrete

multi-group routing and subsequent solid / physics optimization.

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